Module 1 -- Electronic Components Basics

Lesson 1: What is Electricity?

★ Star-1 Beginner ⏰ 40--45 minutes 👤 Age 6--12 years ⚡ 250 XP

Your Mission Today

Welcome, Circuit Explorer! You are about to go on an incredible journey into the invisible world of ELECTRICITY. Your very first mission: make a tiny light glow using nothing but a battery, a wire, and a component called an LED. Are you ready? Let's go!

Learning Objectives

By the end of this lesson, you will be able to:

- Explain electricity as the flow of tiny particles called electrons
- Know what voltage, current, and circuit mean — in your own words
- Successfully light up your very first LED
- Understand why a complete loop (circuit) is needed

What You Need

Item	Qty
9V battery	1
9V battery clip with wires	1
LED (any color)	1
330-ohm resistor	1
Jumper wires	2
Breadboard	1
A piece of wire (or foil strip)	1

How to Teach This Lesson

Step 1: Hook — The Magic Light (5 min)

Before explaining anything, hand the kid a battery, an LED, and a wire.

Say:

> 💬 "Can you make this LED light up? You can only use what is in your hand."

Let them experiment. They might:

- Touch the LED legs to the battery terminals (works! but not safe long-term without a resistor)
- Get confused about which leg is which

The goal is curiosity first. After a minute or two, guide them gently.

> 💬 "You just made electricity flow! Let us figure out why that worked — and how to do it safely."

Award: **+20 XP** **for trying the experiment, even if it did not work the first time!**

Step 2: The Water Analogy (8 min)

Draw this on paper or a whiteboard together:

```
 +-----------------------------+
 |                             |
[Pump]  --water-->  [Narrow pipe]  -->  [Water wheel]
 |                             |
 +-----------------------------+

 Pump        = Battery
 Water       = Electrons (electricity)
 Water pressure = Voltage (V)
 Water flow rate = Current (A)
 Narrow pipe = Resistor (slows flow)
 Water wheel = LED / component doing work
 Full loop   = Circuit
```

Ask the kids:

- "What happens if we cut the pipe?" (Water stops. Circuit is broken. LED goes off.)
- "What if we make the pipe really narrow?" (Less water flows. Less current. Dimmer LED.)
- "What if we have a really powerful pump?" (More pressure. More current. Brighter LED... or it breaks!)

Key words to introduce:

Word	Kid-Friendly Definition
Voltage (V)	How hard the battery "pushes" electricity
Current (A)	How much electricity is actually flowing
Resistance (ohms)	Something that slows the electricity down
Circuit	A complete loop — electricity must go in a circle

Award: **+20 XP** **for answering at least 2 of the 3 questions!**

Step 3: The Circuit on a Breadboard (15 min)

First, explain the breadboard:

```
Breadboard top view:

+ + + + + + + + + +    <-- Power rail (+ connected along the whole row)
- - - - - - - - - -    <-- Ground rail (- connected along the whole row)

a b c d e   f g h i j
[ ][ ][ ][ ][ ]   [ ][ ][ ][ ][ ]   <-- Row 1 (a through e connected together)
[ ][ ][ ][ ][ ]   [ ][ ][ ][ ][ ]   <-- Row 2
[ ][ ][ ][ ][ ]   [ ][ ][ ][ ][ ]   <-- Row 3
...
```

Say:

💬

> "The breadboard is like a pegboard where wires and components snap in. Connected holes share electricity — like water pipes that are joined."

Build the circuit together, step by step:

```
Circuit diagram:

  9V (+) ---- [330-ohm resistor] ---- LED (+leg) ---- LED (-leg) ---- 9V (-)

  On breadboard:
  1. Battery clip: red wire to + rail, black wire to - rail
  2. Resistor: one leg in + rail, other leg in row 5
  3. LED: long leg (anode +) in row 5, short leg (cathode -) in row 7
  4. Wire: row 7 to - rail
```

Go slowly. Have the kid place each component themselves. Describe what you are doing as you go:

> "This resistor is like the narrow pipe — it protects the LED from getting too much electricity."

Power it on together!

When the LED lights up: celebrate! High five! Take a photo.

Award: **+50 XP** **for building your first circuit!**

Step 4: Experiment Time (10 min)

Now let them explore:

Experiment A — Break the Circuit

> "Pull out the resistor. What happens?"

(LED goes off. Circuit is broken.)

Experiment B — Reverse the LED

> "Flip the LED around. What happens?"

(LED goes off. Electricity can only flow one way through an LED!)

> 💬
> "This is called **polarity**. The LED is picky about which direction electricity flows."

Experiment C — The Missing Wire Remove one jumper wire.

> 💬
> "Why did it stop working?"

(The loop is broken. Electricity has nowhere to go.)

Experiment D — Touch the wire to complete the circuit Remove one wire and have the kid hold both bare ends together with their fingers.

> 💬
> "You ARE part of the circuit right now! Electricity is flowing through you (a tiny, tiny, completely safe amount)."

Award: **+30 XP** **for completing at least 3 of the 4 experiments!**

Step 5: Sneak Peek — The Magic Measurement Wand (2 min)

If you have a multimeter handy, hold it up:

> 💬
> "Next lesson, you are going to meet this. It is called a multimeter, but we call it the **Magic Measurement Wand**. It can SEE invisible electricity. It can tell you how strong a battery is, how much a resistor blocks, and whether electricity can flow through ANYTHING."

> 💬
> "Want to know if your pencil conducts electricity? This Wand will tell you. Want to know if your battery is dead? This Wand knows. Get excited — it is your new best friend!"

Award: **+10 XP** **just for being curious about the Wand!**

Step 6: Wrap Up and Quiz (5 min)

Question 1: "What does a battery do?" (Pushes electricity around the circuit) **+20 XP** **for a correct answer!**

Question 2: "Why do we need a complete loop?" (So electricity has somewhere to go) **+20 XP** **for a correct answer!**

Question 3: "What happens if I remove the resistor?" (Too much current — LED could burn out) **+20 XP** **for a correct answer!**

Bonus question for older kids (10+):

> "If the battery is 9V and the LED needs 2V, where does the extra 7V go?"

(Into the resistor — it "uses up" the leftover voltage as heat) **+30 XP** **bonus!**

Fun Extras (Optional)

Make It Glow in the Dark

Turn off the lights and show off the glowing LED. Kids love this!

The Lemon Battery

Stick a copper coin and a zinc nail into a lemon. Can it light the LED?

> "A lemon is a battery! Sour = chemistry = electricity. In our next lesson, you will learn to use the Magic Measurement Wand to measure exactly how many volts a lemon makes!"

Lesson 1 Complete!

```
==============================================

   CIRCUIT STARTER BADGE UNLOCKED!

   Skills unlocked:
   [check] Know what voltage, current, and resistance mean
   [check] Built your first LED circuit
   [check] Understand what a circuit loop is

==============================================
```

XP Breakdown:

Activity	XP
Hook experiment	20
Water analogy questions	20
Build first circuit	50
Experiments (3 of 4)	30
Wand sneak peek	10
Quiz (3 questions)	60
Bonus question	30
TOTAL POSSIBLE	**220**

Coming Up Next...

In **Lesson 2**, you will meet your **Magic Measurement Wand** — the multimeter! You will discover its three superpowers and use it to test whether coins, pencils, and even your own finger can conduct electricity. Get ready!

Troubleshooting

Problem	Likely Cause	Fix
LED does not light up	LED reversed	Flip the LED around
LED does not light up	Bad connection	Check each wire in the breadboard
LED flickers	Loose wire	Press wires firmly into the breadboard
LED very dim	Wrong resistor	Try a smaller resistor (e.g., 100 ohms)
Nothing works	Battery dead	You will learn to test this with the Wand in Lesson 2!

Parent/Instructor Notes

- For ages 6—8: Skip the bonus question. Focus entirely on the hands-on experience. The key win is "I made a light turn on!"
- For ages 9—12: Encourage them to explain WHY the LED goes off when they flip it (polarity concept).
- The "Sneak Peek" of the multimeter at the end builds anticipation for Lesson 2 and sets up the "Magic Measurement Wand" framing.
- Track XP on paper or a whiteboard. Kids respond incredibly well to seeing their score go up. Consider making a poster-sized XP tracker for the wall.